Git Detailed Concepts Guide

This document explains in a clear and structured manner the following Git concepts: 1. git remove command 2. git log command 3. Drag and drop upload in GitHub 4. Commit hash

Each topic is explained with definition, purpose, and step by step usage.

git
remove

1.

Correct command name is:

```
git rm
```

Definition

git rm is used to remove a file from both your working directory and the Git tracking

system. When to use git rm

Use git rm when: - You want to delete a file permanently - You want Git to stop tracking a

file Step by Step Example

Step 1: Check existing files

```
git status
```

Step 2: Remove a file

```
git rm file.c
```

This command: - Deletes the file from your folder - Stages the deletion automatically

Step 3: Commit the deletion

```
git commit -m "Removed file.c"
```

Step 4: Push to GitHub

```
git push
```

Now the file will also be removed from GitHub.

Important Variation

If you want to remove file only from Git tracking but keep it in folder:

```
git rm --cached file.c
```

This keeps the file locally but removes it from repository tracking.

git log

1.

Definition

git log shows the history of commits made in the repository. It

displays: - Commit hash - Author name - Date - Commit message

Basic Command

```
git log
```

Output Example Structure

```
commit a1b2c3d4e5f6...
Author: Name
Date: Time

Commit message
```

Useful Variations

Show short version:

```
git log --oneline
```

Shows each commit in one line with short hash.

Show limited number of commits:

```
git log -n 3
```

Shows last 3 commits.

Why git log is important

- To check what changes were made
- To copy commit hash
- To track project history

1. Drag and Drop in GitHub

Definition

Drag and drop is a method of uploading files directly to GitHub using the browser without using Git commands.

When to use it

- When Git is not installed
- When submitting assignments quickly
- When making small changes

Step by Step Process

Step 1: Login to GitHub

Step 2: Open repository

Step 3: Click Add file

Step 4: Click Upload files

Step 5: Drag file from your system into browser window

Step 6: Write commit message

Step 7: Click Commit changes

This creates a commit directly on GitHub.

Important Note

Drag and drop does not use your local Git repository. It creates commits online only.

Commit Hash

1.

Definition

A commit hash is a unique identification number generated by Git for every commit. It is a long string of letters and numbers.

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Example:

```
a1b2c3d4e5f67890abcd1234ef567890abcd12345
```

Purpose of Commit Hash

- Identifies a specific version of project
- Used for reverting changes
- Used for checking specific commits
- Used in pull requests and comparisons

How to View Commit Hash

```
git log
```

Or short version:

```
git log --oneline
```

Using Commit Hash

Checkout a specific commit:

```
git checkout <commit-hash>
```

Revert a commit:

```
git revert <commit-hash>
```

Important Understanding

Every time you commit, Git generates a new unique hash. No two commits have the same

hash. Summary

- git rm removes files from repository
- git log shows commit history
- Drag and drop uploads files directly on GitHub website
- Commit hash uniquely identifies each commit

If these commands are practiced step by step, understanding of Git version control becomes much stronger.